

Assignment 05 – The HTML Web Table Extractor

1. Module Name

Module 1 – Automation with Selenium

2. Experiment Title

The HTML Web Table Extractor

3. Problem Statement

Create a Selenium Python automation script to locate a web table, identify its rows and columns, extract the table data, and search for a specific author. The script should retrieve the Book Name, Author, Subject, and Price for matching records.

4. Objective

The objective of this experiment is to learn how to work with HTML web tables using Selenium WebDriver. The automation should locate the BookTable, count rows and columns, display the headers and table data, and retrieve the price of books written by the specified author.

5. Tools, Software and Concepts Used

- Python 3
- Selenium WebDriver
- Google Chrome and ChromeDriver
- VS Code
- WebDriverWait and Expected Conditions
- XPath locators
- find_elements()
- HTML table rows and columns
- Loops and conditional statements
- Assertions for result verification

6. Source Code / Implementation Steps

1. Open the Automation Testing Practice website: <https://testautomationpractice.blogspot.com/>.
2. Maximize the browser window.
3. Use WebDriverWait to wait until the BookTable is visible.
4. Locate the table using the XPath `//table[@name='BookTable']`.
5. Find all table rows and header cells.
6. Display the number of rows and columns and print the table headers.
7. Iterate through the table rows and extract each column value.
8. Search for records where the author is **Mukesh**.
9. Retrieve and display the Book Name, Author, Subject, and Price for matching records.
10. Use an assertion to verify that at least one matching record is found.
11. Keep the browser open temporarily so that the required screenshots can be captured.

7. Output / Labelled Screenshots

The following screenshots are captured during execution and stored in the **Source_Code/Screenshots/** folder:

- Screenshot 1 – Web Table displayed in the browser
- Screenshot 2 – Extracted table data and matching Mukesh records
- Screenshot 3 – Successful execution message in the terminal

8. Result

The Selenium script successfully located the BookTable on the Automation Testing Practice website, read the table headers and records, searched for books written by Mukesh, and retrieved the corresponding book details and prices. The assertion confirmed that a matching record was found. The experiment was completed successfully.

9. Observation

The experiment demonstrated that Selenium can interact with HTML tables by locating the table element and then iterating through its rows and cells. Using a specific XPath for the BookTable made the element identification more reliable. The extracted values can be processed using normal Python loops and conditional statements.

10. Conclusion

This experiment provided practical experience in extracting and processing data from HTML web tables using Selenium Python. I learned how to locate a table, identify its rows and columns, extract cell values, search for a particular record, and verify the result using an assertion. The assignment successfully demonstrated web table automation using Selenium.